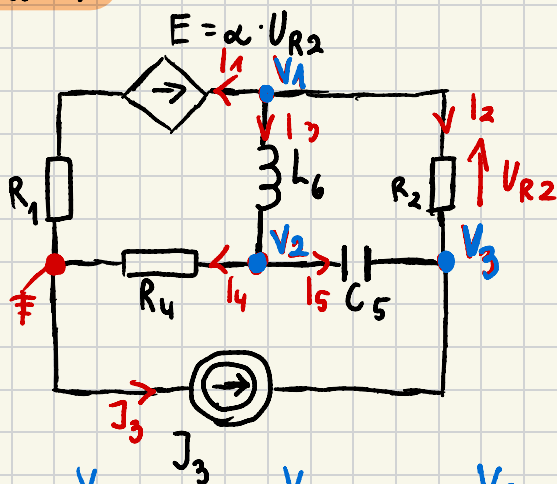


ĆWICZENIA 3

Zad 1



$$\begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} \begin{bmatrix} Y_1 + Y_6 + Y_2 & -Y_6 & -Y_2 \\ -Y_6 & Y_4 + Y_6 + Y_5 & -Y_5 \\ -Y_2 & -Y_5 & Y_2 + Y_5 \end{bmatrix} \cdot \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} E \cdot Y_1 \\ 0 \\ J_3 \end{bmatrix}$$

$\alpha Y_1 V_1 - \alpha Y_1 V_3$

Wyznaczamy wzory na "wiadome"

$$Z_1 = R_1 \Rightarrow Y_1 = \frac{1}{R_1}$$

$$Z_2 = R_2 \Rightarrow Y_2 = \frac{1}{R_2}$$

$$Z_5 = -jX_C \Rightarrow Y_5 = jX_C$$

$$Z_6 = jX_L \Rightarrow Y_6 = -jX_L$$

$$E = \alpha \cdot U_{R2} = \alpha (V_1 - V_3)$$

$$\begin{bmatrix} Y_1 + Y_6 + Y_2 - \alpha Y_1 & -Y_6 & -Y_2 + \alpha Y_1 \\ -Y_6 & Y_4 + Y_6 + Y_5 & -Y_5 \\ -Y_2 & -Y_5 & Y_2 + Y_5 \end{bmatrix} \cdot \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ J_3 \end{bmatrix}$$

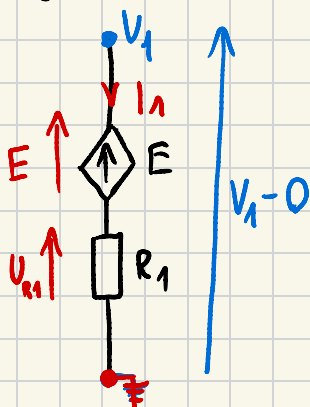
PRZEKSZTAŁCENIE ZACHODZĄCE W PAMIĘCI

$$\begin{bmatrix} \alpha Y_1 V_1 - \alpha Y_1 V_3 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \alpha Y_1 & 0 & -\alpha Y_1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix}$$

W kalkulator naukowy albo metodą wyznacznikową

Dostajemy: V_1, V_2, V_3

Wyznaczam U_{R1}



$$V_1 - 0 = E + U_{R1}$$

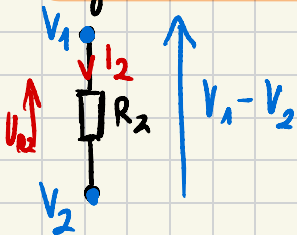
$$V_1 = E + U_{R1}$$

$$U_{R1} = V_1 - E$$

$$U_{R1} = V_1 - \alpha V_1 + \alpha V_2$$

$$I_1 = U_{R1} \cdot Y_1$$

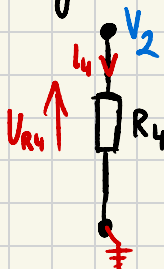
Wyznaczam U_{R2}



$$U_{R2} = V_1 - V_2$$

$$I_2 = U_{R2} \cdot Y_2$$

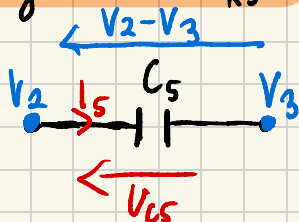
Wyznaczam U_{R4}



$$U_{R4} = V_2$$

$$I_4 = U_{R4} \cdot Y_4$$

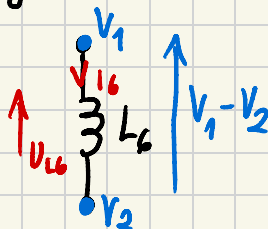
Wyznaczam U_{C5}



$$U_{C5} = V_2 - V_3$$

$$I_5 = U_{C5} \cdot Y_5$$

Wyznaczam U_{L6}



$$U_{L6} = V_1 - V_2$$

$$I_6 = U_{L6} \cdot Y_6$$

Sprawdzamy, czy PPK zachowane!

$$V_0: I_1 + I_4 - I_3 = 0$$

$$V_1: -I_1 - I_6 - I_2 = 0$$

$$V_2: I_6 - I_4 - I_5 = 0$$

$$V_3: I_5 + I_2 + I_3 = 0$$

Obliczamy moc pozorną zespoloną

$$\underline{S}_1 = \underline{U}_{R1} \cdot \underline{I}_1^*$$

$$\underline{S}_E = \underline{E} \cdot \underline{I}_1^*$$

$$\underline{S}_2 = \underline{U}_{R2} \cdot \underline{I}_2^*$$

$$\underline{S}_3 = (0 - \underline{V}_3) \cdot \underline{I}_3^*$$

$$\underline{S}_4 = \underline{U}_{R4} \cdot \underline{I}_4^*$$

$$\underline{S}_5 = \underline{U}_{C5} \cdot \underline{I}_5^*$$

$$\underline{S}_6 = \underline{U}_{L6} \cdot \underline{I}_6^*$$

$$C = 1 + 2j$$

$$C^* = 1 - 2j$$



Bilans mocy

$$\underline{S}_1 + \underline{S}_E + \underline{S}_2 + \underline{S}_3 + \underline{S}_4 + \underline{S}_5 + \underline{S}_6 = 0$$

$$R_1 = 10$$

$$R_2 = 15$$

$$R_3 = 10$$

$$C_5 = 0,0005$$

$$L_6 = 0,15$$

$$\omega = 100 \frac{\text{rad}}{\text{s}}$$